Checkpoint D — Architecture and Changeability

QuantRule

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1 Checkpoint D — Architecture and Changeability

1.1 Claim

The local-first architecture keeps the deterministic engine isolated from the desktop UI and the distribution channel, so a model, strategy, or UI change stays local and testable.

1.2 Current architecture

- One local quantum: Tauri UI + application service + pure engines.
- Pipes-and-filters inside a modular monolith.
- CLI and UI are adapters over the same application service.
- The download page is a static distribution channel.
- The research memory is a SQLite store behind the application service.

1.3 Change scenarios

Change	Boundary	Evidence
New model reading	`crates/quantrule-model`	golden, causality, determinism tests
New strategy rule	`crates/quantrule-backtest`	strategy boundary tests
UI screen	`apps/desktop/src`	unit and browser E2E
New command	`crates/quantrule-shell`	command tests and help
New distribution channel	`docs/architecture/deployment.md`	clean install and recorded digest
Team cloud (future)	new control plane	not implemented; new ADR

1.4 Evidence

- Crate boundaries and structural checks.
- Model/strategy boundary tests.
- The Tauri UI calls `quantrule-app` through typed commands; it does not import an engine directly.
- The Phase 3 verification report records the clean install, offline launch, and memory round-trips.

1.5 Limit

The UI does not run backtests or analytics; those remain CLI operations. The team cloud is a future quantum and is not covered by this evidence.

1.6 Decision

Keep the pipeline/modular-monolith style; do not add a server or microservices for the solo increment.